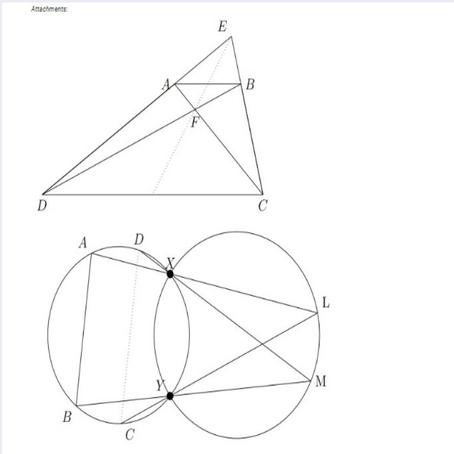
INDIAN INSTITUTE OF INFORMATION TECHNOLOGY RAICHUR
Mathematics Club
PROBLEM NUMBER: WP2609
DATE: MAY 1, 2026
DUE DATE: MAY 10, 2026

WEEKLY PROBLEM

EXERCISE.



1. **Part 1**

   Two circles G1, G2 interest at points X,Y. Choose two other points A,B on G1 as shown in the figure. The line segment from A to X is extended to intersect G2 at point L. The line segment from A to X is extended to meet G1 at point C. Likewise the line segment from M to X is extended to meet G1 at point D. Show that AB is parallel to CD.

2. **Part 2**

   A triangle CDE is given. A point A is chosen between D and E. A point B is chosen between C and E so that AB is parallel to CD. Let F denote the point of intersection of segments AC and BD. Assume that the trapezium ABCD is a cyclic quadrilateral. Show that the line joining E and F bisects both segment AB and segment CD.

3. **Part 3**

   Let G1 and G2 be two circles intersecting at points X and Y. Using the results of parts (i) and (ii), show that the center of each circle can be determined as the intersection of explicitly constructed lines, where each line is defined by two points that arise from the intersection of lines and circles constructed in parts (i) and (ii).

(Scan this to submit solution!)

Top Submissions for last Week's Problem (WP2608):

1. **RUTHALA JASWANTH KUMAR** - MC24B1036
2. **RONGALA TEJA ABHIRAM** - CS25B1033
3. **MANSWI RAHUL PRADHAN** - MC25B1023
4. **ADITYA SHUKLA** - MC24B1002